

Hadoop to EMR Migration Delivery Kit

Overview

Customers can now engage AWS Professional Services to provide technical expertise and delivery best practices as they are migrating on-premises Hadoop workloads to AWS. Hadoop Migration Delivery Kit (HMDK) provides customers with required information to help them structure their execution plan for a migration, including Epics, Stories, high-level Sprint breakdowns, and a collection of Hadoop migration patterns and accelerators. The goal of the Hadoop Migration delivery kit is to build foundational architecture and migration strategy, delivery approach, weekly sprints, initial data on-boarding, and roles and responsibilities identified for the migration effort.

For the introduction to Engagement Delivery Framework (EDF) for Hadoop migrations, refer to the EDF presentation below for an overview of the business outcomes, migration phases, target architecture and accelerators, and project governance.

Scope

The Hadoop Migration Delivery Kit (HMDK) is intended to do the following:

1. Define target state architecture (with some draw.io diagrams for your usage and editing)
2. Define the data migration strategy
3. Define metadata migration strategy
4. Security Policies Migration strategy
5. Define data governance strategy
 1. Establish Central Data Lake
 2. Enable AWS services such as AWS Lake Formation
 3. Standardize data producer and consumer relationships between AWS accounts / environments
 4. Define Table registration processes
 5. Define Table subscription processes
6. Define Job scheduling strategies
7. Establish standards on EMR cluster sizing and configurations
8. Define code migration and sanity testing strategies. For example: Hortonworks Data Platform version to Amazon EMR version
9. Code migration - Automatic conversion of orchestration scripts:
 1. Convert Oozie workflows to Airflow DAGs
 2. Convert Airflow DAGs to MWAA DAGs
10. Define project governance aka Amazon Engagement Delivery Framework
11. Data validation and cutover strategy

Pre-requisites

This EDF assumes the following prerequisites are complete. If any of the assumptions are not true, these are gaps in the plan and estimates.

1. An EMR Migration approach has been defined as part of EMR Migration Align Phase. If you have not executed this offering, complete this first, [On Premise Hadoop Migration to Amazon EMR Discovery](#).
2. A Data Lake design has been completed as part of the Align Phase, and Data Lake development is complete or in progress.
3. A security assessment has been completed.
4. A landing zone has been established.
5. DevOps tooling has been selected.

What's Included in This Delivery Kit

- Hadoop to EMR Migration Approaches and Accelerators
- Data Migration
- Metadata Migration Stories
- Security Policies Migration Stories
- Security Stories
- Code & Job Migration
- Testing
- Operation Stories

Resourcing

Once the Align phase is complete, the Launch phase can be delivered by AWS ProServe Consultants and Partner's. When Partner's are involved, there will still need to be a core set of AWS Consultants who own certain workstreams, and are available to provide technical best practices and oversight. See the below table for details and scope for each role.

Role	Scope	Timeframe
Program Level		

Sr. Data Architect (L6)	Project lead who defines target state architecture. Responsibilities include the following: • Define Amazon EMR Cluster provisioning and sizing strategies • Define EMR networking strategy • Define data migration strategy • Define Hive Metastore migration strategy • Define Job scheduling and orchestration strategies • Define end-to-end validation strategy • Enable Glue Data Catalog and Lake Formation services • Define CI/CD strategy • Prioritize Hadoop jobs in collaboration with customer and data engineering team	Full engagement, reduce to part time after 6-8 weeks.
Sr. Engagement Manager (L6)	Engagement Managers lead individuals, small teams, or groups of teams and play a leadership role in scoping requirements with customers, analyzing project goals, creating a plan to achieve those goals: • Program Governance and oversight • Resource management • Financial management • Develops and manages program governance plans, program schedules, and program milestones	Full engagement, full time
Sr. Security Consultant (L6)	• Identify and document security requirements per AWS best practices • Develops or documents existing tagging strategy • Conducts security reviews and identifies tasks which require remediation • Determines IAM Roles/Policies (in absence of CIA)	4-6 weeks at beginning of engagement, full time
Sr. DevOps Architect (L6)	• Develop architecture for CICD pipelines • Code repos creation • Develops the CICD strategy • Infrastructure automation and build out	4-6 weeks at beginning of engagement, full time

EMR Specialists (L6, L7)	Professional Services Consultant specialized in Amazon EMR. Responsibilities include the following: • Customizing EMR with custom AMIs • Customizing EMR launch configuration and installing custom applications part of EMR runtime • EMR cluster tuning • Tuning job performance issues • Assisting data engineering team and data architect and solving technical blockers	Full engagement, part time
Pod Level		
Product Owner (Customer)	• Define job prioritization for migration for workstream • SME for business logic and job dependencies • Pre-cutover validation and cutover approval • Support for data validation	
Scrum Master (Customer, AWS)	• Facilitate recurring Agile ceremonies (sprint planning, standup, refinement, sprint review) • Sprint reporting	
Data Engineers (L4, L5, Partner) 3-4 per pod	Data Engineers are responsible for execution of workload migration. Responsibilities include the following: • Job dependency analysis • Smoke testing of EMR jobs with data and without data • Porting Hadoop jobs to EMR • Converting orchestration scripts (Oozie, Airflow) scripts to EMR preferred orchestration scripts (Step Functions, Amazon MWAA) • Populate Glue Data Catalog • End to end execution of Jobs on EMR and data validation • Job cutover • Prepare Developer Guide(s) • Prepare relevant project documentation	Full engagement, full time

QE Engineers (L4, Partner) 1 per pod	Quality Engineers (QE) are responsible for validation of data once workload has is migrated. Responsibilities include the following: • Perform data validation • Create automated framework for completing data validation	Full engagement, full time
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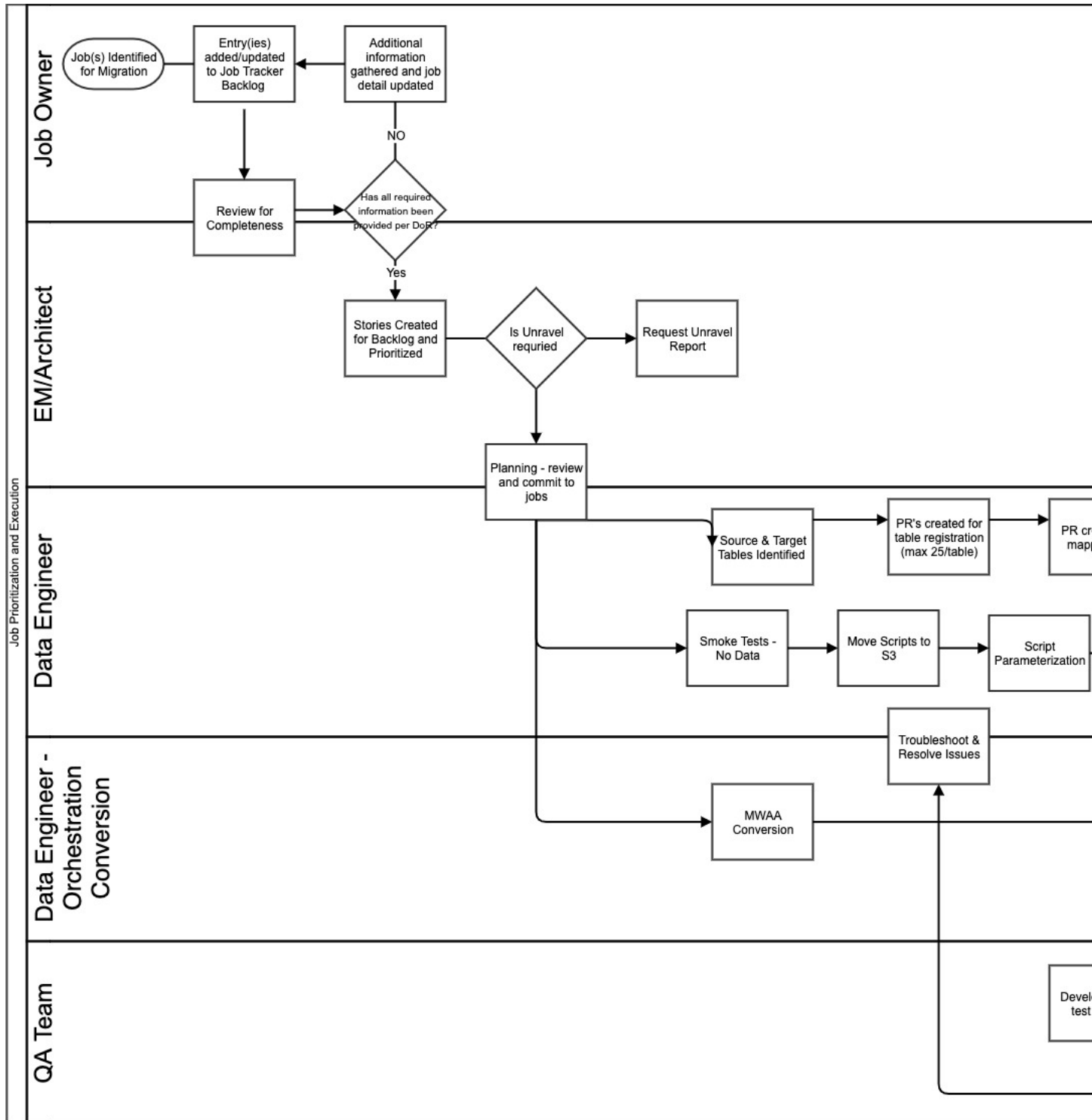
Assumptions

- The Offering targets customers running on-prem Hadoop workloads (Cloudera, HortonWorks, MAPR, open-source).
- Customer must have dedicated resources (Data/Solution Architect, Data Engineer, ETL Specialist, Migration Lead) during the migration to support the analysis of current state environment and the desired future state outcomes.

Audience

Migrating a job from on-premises Hadoop to EMR requires a set of steps to be completed and ALWAYS starts and ends with the Customer. To ensure consistency for each job migrated, having a standard set of stories is critical to maintain a constant velocity and delivery cadence.

Big Diagram (Scroll to the right to see full).



Amazon EDF User Stories

Initiation and management of a Hadoop Migration utilizes the standard approach used within AWS ProServe for establishing the delivery team and governance structure. Following this pattern will ensure a standardized approach and clear pattern for achieving business outcomes.

Story	Goal	List of Artifacts	Artifact Link
Onboard resources	As an Engagement Manager, I would like to allocate and onboard resources to support the Data Lake engagement to support the business with their analytical vision and business value.	- Salesforce Resource allocation request. - Resource allocated and assigned for the duration of this engagement. Hardware and software required for the engagement is identified and procured.	Hadoop-Migration-Job-Description
Identity Partner resources	As an Engagement Manager, we have the team ensure the partner resources have the right skills.	Partner Work Order executed. Partners on-boarded.	Hadoop-Migration-Job-Description
Introduction to EDF & customer training	As an Engagement Manager, I would like to work with the customer to identify a project management methodology, preferably EDF, so that we can efficiently and effectively manage the project.	EDF training plan or customer & internal resource. EDF training completed.	EDF for Hadoop Migrations: EDF for Hadoop to EMR Migrations Customer Training EDF+Training
Develop an engagement charter	As an Engagement Manager, I would like to establish a project sponsor so all team members are in alignment on project objectives, scope, and metrics to measure success. Please note that the scope is only limited to the engagement or MVP.	Approved Charter Business Case	

Delivery Quality Review	As an Engagement Manager, I need to ensure that my project is delivering on time and on schedule with the highest quality.	Delivery Quality Review complete.	Analytics_Delivery_Review
Define roles & responsibilities	As a Project Sponsor, I would like AWS, the partner and customer to understand their roles and responsibilities so that all parties are in alignment (RACI).	A clearly defined RACI is communicated to all parties to sign off and to confirm they are in alignment. Well-defined organization model with clear roles defined.	Link For Sample RACI, Roles & Responsibilities
Create Lake Formation Migration Plan	As a Project Sponsor, I would like to create a plan to migrate to Lake formation from our current environment with Glue, Athena, S3	Lake Formation migration sprint plan with clear outcomes.	AWS Lake Formation Catalog Migration Plan
Create a project schedule with key dates and milestones	As an Engagement Manager, I would like to create a project schedule so I can manage key dates and milestones.	Given that all key milestones are identified in a project schedule then the schedule should be communicated and measured weekly. Initial Planning for first few sprints with key milestones.	
Prepare an engagement budget	As an Engagement Manager, I would like to create an engagement budget So I can ensure the	Engagement Financial Details with Weekly breakdown.	

	budget plan matches the SOW.	Approval from stakeholder on the financial reporting model.	
Establish an engagement management process	As an Engagement Manager, I would like to establish an engagement management approach and tools to better facilitate the project.	Given agile project methodology, when the project kicked off, then all engagement management processes should be in place including onsite/offsite schedule, resource access.	
Conduct an engagement kickoff	As an Engagement Manager, I would like all parties to participate in a project kickoff meeting so we can share information and determine how to work together.	An engagement kickoff meeting is completed shortly after the SOW is signed and the customer is ready to initiate the engagement.	
Start a risk register	As an Engagement Manager, we have compared our delivery excellence against the well architected review to ensure there are no gaps.	Identification of any risks from the well architected review	
Run Well Architected review Analytic Lens	As an Engagement Manager, we have compared our delivery excellence against the well architected review to ensure there are no gaps	Identification of any risks from the well architected review	Data Analytics Lens
Success metrics clearly documented	As an engagement manager success of the project is clearly document in a 1 slider, so that when closing the project we are exceeding expectations	1 slider created with documented success metrics 1 slider reviewed and approved by project stakeholders	ProServe Book of Projects

